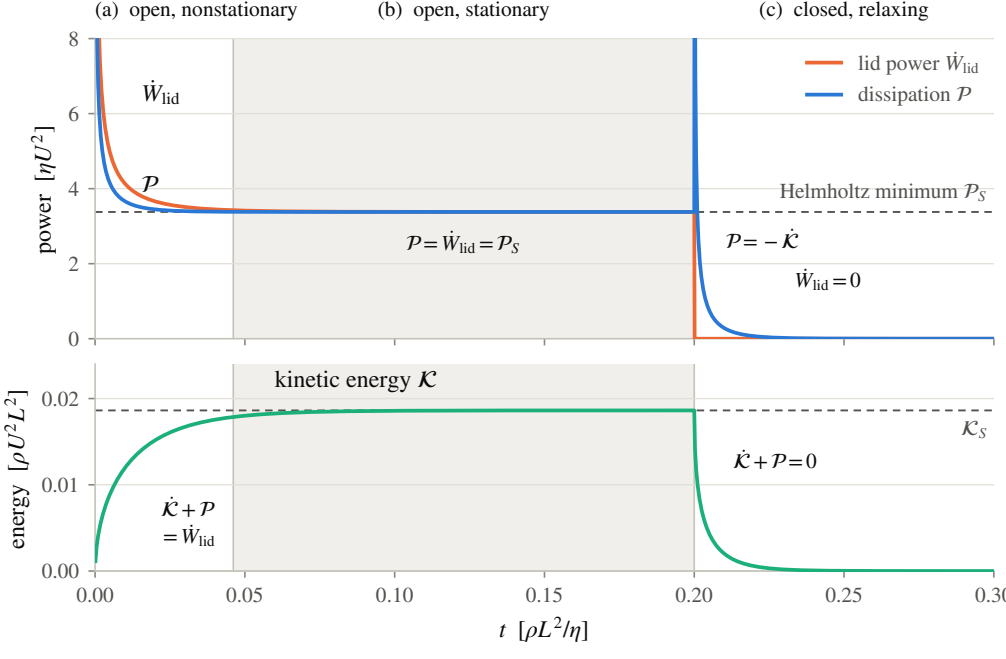
(a) open, nonstationary
(b) open, stationary
(c) closed, relaxing
lid power $\dot{W}_{\mathrm{lid}}$
dissipation $\mathcal{P}$
$\dot{W}_{\mathrm{lid}}$
$\mathcal{P}$
Helmholtz minimum $\mathcal{P}_S$
$\mathcal{P} = \dot{W}_{\mathrm{lid}} = \mathcal{P}_S$
$\mathcal{P} = -\dot{\mathcal{K}}$
$\dot{W}_{\mathrm{lid}} = 0$
power $[\eta U^2]$
8
6
4
2
0
kinetic energy $\mathcal{K}$
$\mathcal{K}_S$
$\dot{\mathcal{K}} + \mathcal{P} = 0$
$\dot{\mathcal{K}} + \mathcal{P} = \dot{W}_{\mathrm{lid}}$
energy $[\rho U^2 L^2]$
0.02
0.01
0.00
0.00
0.05
0.10
0.15
0.20
0.25
0.30
t $[\rho L^2/\eta]$